Lecture 9 – Smooth Manifolds

Tutorial Exercises

4 November 2026

1 Charts and Manifolds

1. Show that $\psi_- : (0, 2\pi) \to S^1$ with $\psi_-(r) = (\sin r, -\cos r)$ is a 1-chart and compatible with $\psi(r) = (\sin r, \cos r)$ from example 9.4e on S^1.
2. Find a 1-chart on $\mathbb{R}$ with a non-empty domain that is compatible with the non-standard chart $\varphi\colon \mathbb{R} \to R$ given by $\varphi(r) = r^3$ from example 9.8.
3. Describe what 0-manifolds are.
4. Prove lemma 9.9 (topology on manifold).
5. Show that the topology induced on $\mathbb{R}$ by the base $\{\mathrm{id}_{\mathbb{R}}\}$ is the same as the topology induced by the Euclidean metric.
6. Provide a 2-chart on the complex numbers $\mathbb{C}$.
7. Prove theorem 9.10 (reparameterisation and local determination of charts)
8. Let $\mathcal{A}$ be an atlas on M and $U \subseteq M$ open. Show that there is an atlas $\mathcal{A}_U$ on U.

2 Hausdorff Spaces

1. Show that if $A \subseteq X$, then A with the subspace topology is also Hausdorff.
2. Show that the topology induced by a metric is Hausdorff.
3. Show that if X is a Hausdorff topological space and path-connected, then all points x, y can be connected by in injective path (called an *arc*).
4. Pushout in **Top**: Suppose that $X \xleftarrow{f_1} Z \xrightarrow{f_2} Y$ are continuous maps between topological spaces. Show that there is a topological space $f_1 \coprod f_2$ with continuous maps $X \xrightarrow{k_1} f_1 \coprod f_2 \xleftarrow{k_2} Y$ that renders the diagram on the left commutative, and such that for all topological spaces V and continuous maps as in the diagram on

right there is a unique continuous map $g\colon f_1 \coprod f_2 \to V$ with $g \circ k_i = g_i$. The triple $(f_1 \coprod f_2, k_1, k_2)$ is called the **pushout** of f_1 and f_2.

$$\begin{array}{ccc} Z & \xrightarrow{f_1} & X \\ {\scriptstyle f_2}\downarrow & & \downarrow{\scriptstyle k_1} \\ Y & \xrightarrow{k_2} & f_1 \coprod f_2 \end{array} \qquad \begin{array}{ccc} Z & \xrightarrow{f_1} & X \\ {\scriptstyle f_2}\downarrow & & \downarrow{\scriptstyle g_1} \\ Y & \xrightarrow{g_2} & V \end{array}$$

5. Let $\mathbb{R} \xleftarrow{i_1} \mathbb{R} \setminus \{0\} \xrightarrow{i_2} \mathbb{R}$ be the inclusion maps. The **line with two origins** or **bug-eyed line** $\mathbb{R}^{\text{oo}}$ is the pushout $i_1 \coprod i_2$. Show that there is a smooth atlas on $\mathbb{R}^{\text{oo}}$ but that it is not Hausdorff (use item 3).

3 Countable Bases

- Show that $\mathbb{R}$ is 2nd countable.
- Let X be 2nd countable and $A \subseteq X$. Show that A with the subspace topology is also 2nd countable.
- Show that if $(M, \mathcal{A})$ is a manifold and $U \subseteq M$ open, then $(U, \mathcal{A}_U)$ a manifold (see above for the subspace atlas $\mathcal{A}_U$).
- Let $GL(n)$ be the set of invertible matrices on $\mathbb{R}^n$. Show that $GL(n)$ can be equipped with an atlas that makes it a manifold. *Hint: Use show that $GL(n)$ is open in the space of all $n \times n$ matrices.*

4 Some Sources

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